

# PLANRS ACADEMY

## Patterns & Sequences: Discovering the World of Patterns

|                    |                                                 |                |                                                   |
|--------------------|-------------------------------------------------|----------------|---------------------------------------------------|
| <b>Grade</b>       | 2 (ages 7-8)                                    | <b>Subject</b> | Mathematics (Algebra Readiness)                   |
| <b>Duration</b>    | 60 minutes (can split into 2 x 30-min sessions) | <b>Medium</b>  | English with Hindi / Tamil phrases for engagement |
| <b>Prepared by</b> | PLANRS Academy                                  |                |                                                   |

This lesson introduces Grade-2 students to the concept of repeating and growing patterns through culturally relevant Indian examples—rangoli, kolam, phulkari embroidery, bangles, and festival diyas. Students will identify, extend, complete, and create their own patterns while building early algebraic thinking.

### 1. Learning Objectives

By the end of this lesson, students will be able to:

- Define what a pattern is and explain it in their own words.
- Identify the pattern core (the repeating unit) in shape, colour, size, and number sequences.
- Classify patterns by core type: AB, AAB, ABB, ABC, AABB.
- Extend a given repeating or growing pattern by at least two more elements.
- Find and fill in missing elements within a pattern.
- Connect patterns to real-life Indian cultural objects (rangoli, saree borders, traffic lights, bangles).
- Describe simple growing patterns using skip counting (by 2s and 5s).

### Curriculum Alignment (NCF 2023 / CBSE Grade 2)

| Strand                                      | Competency                            | Indicator                                                          |
|---------------------------------------------|---------------------------------------|--------------------------------------------------------------------|
| <b>Number &amp; Algebra</b>                 | Recognise patterns                    | Identify and describe repeating units                              |
| <b>Number &amp; Algebra</b>                 | Extend patterns                       | Predict the next term of a sequence                                |
| <b>Number &amp; Algebra / Communication</b> | Skip counting / Mathematical language | Count by 2s, 5s, and 10s; use 'pattern', 'core', 'repeats', 'next' |

## 2. Materials & Resources

| Resource                        | Purpose / Notes                                                                 |
|---------------------------------|---------------------------------------------------------------------------------|
| <b>Presentation Slides</b>      | 21-slide deck (PDF provided) with Indian characters, rangoli, and kolam visuals |
| <b>Whiteboard / Blackboard</b>  | For live demonstrations and student responses                                   |
| <b>Colour Tiles or Blocks</b>   | Red, green, blue blocks for hands-on pattern making                             |
| <b>Bead Necklace Props</b>      | Thread + beads of 3 colours—one per pair (optional but highly effective)        |
| <b>Pattern Worksheet</b>        | Printable sheet with 'what comes next?' and 'find the missing piece' tasks      |
| <b>Shape Stamps / Stickers</b>  | Circles, squares, triangles for creating own patterns                           |
| <b>100-Chart</b>                | Laminated, for skip-counting number patterns                                    |
| <b>Rupee Coins (play money)</b> | For skip-by-5 counting pattern (5, 10, 15, ...)                                 |
| <b>Sticky Notes (3 colours)</b> | For the 'Pattern Hunt' gallery activity                                         |

## 3. Lesson Structure (60-Minute Plan)

| Time / Slides                | Teacher Actions                                                                                                                                                                                                                                                                                                                                                 | Student Actions                                                                                          |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| <b>0–8 min</b><br>Slides 1–2 | <b>Warm-Up / Hook:</b> Display Slide 1 (Rangoli cover image). Ask: "Have you seen rangoli at home or a festival? What do you notice about the design?" Show Slide 2 character Priya says " <i>Yeh pattern baar baar aata hai!</i> " Clap a simple AB rhythm (clap, pat, clap, pat...) and invite students to join. Ask: "What would come next if I kept going?" | Observe rangoli image and share observations. Join the clap-pat rhythm pattern. Predict what comes next. |

| Time / Slides             | Teacher Actions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Student Actions                                                                                                                                                         |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8–18 min<br>Slides 3–5    | <b>Concept Introduction - What Is a Pattern?</b><br>Show Slide 3. Define: "A pattern is something that repeats in the same order, again and again." Give 3 examples: saree border, traffic lights, colourful bangles. Show Slide 4 - Introduce "pattern core": the part that repeats. Demonstrate on whiteboard. Draw shapes and ask "What is the core?". Show Slide 5 - Types of cores: AB, AAB, ABB, ABC, AABB with symbols. Point to Kolam and Phulkari embroidery as real AB/AABB examples. | Listen and repeat the definition. Identify examples from their own life. Point to the core in each teacher-drawn pattern. Classify the type (AB/AAB) when teacher asks. |
| 18–24 min<br>Slide 6      | <b>Patterns in Everyday Life:</b> Show Slide 6—real-life examples (beaded jewellery, blouse border, chapatis, traffic lights). Ask: "Look around our classroom—can anyone spot a pattern?" Invite 2–3 students to point out patterns on clothing or surroundings. Explain size patterns: big-small-big-small on a plate of chapatis.                                                                                                                                                            | Observe and identify patterns in the classroom. Share one pattern they see at home.                                                                                     |
| 24–34 min<br>Slides 7–8   | <b>Extending Patterns (What Comes Next?):</b><br>Show Slide 7. Teach the 3-step method: (1) Find the core, (2) Count repetitions, (3) Add next in order. Model with diya example: Small, Big, Small, Big → answer: Small, Big. Show Slide 8 peacock-flower pattern; ask students to add next two elements. Pair activity (3 min): Give colour tiles; one student makes pattern, partner extends it.                                                                                             | Follow the 3-step method with teacher. Predict and answer for diya and peacock examples. Make and extend partner's colour-tile pattern.                                 |
| 34–42 min<br>Slides 9–10  | <b>Missing Elements in a Pattern:</b> Show Slide 9 - Strategy: look at what comes before AND after the blank. Model bead necklace example with props if available. Show Slide 10 —'Pattern Detective Priya': red, green, blue, red, green → blue. Give worksheet Task 2 (missing elements); circulate and give hints.                                                                                                                                                                           | Apply the 'look before and after' strategy. Answer Priya's necklace puzzle. Complete worksheet Task 2 individually.                                                     |
| 42–50 min<br>Slides 11–16 | <b>Mini Quiz (Interactive):</b> Show Slides 11–16 (3 questions with answers). Q1: Identify the pattern core (flower + leaf). Q2: Extend sequences. Q3: Missing element in bead sequence. Use thumbs-up/thumbs-down for quick formative check. Praise with <i>"Well done! Bilkul sahi!"</i> after each answer.                                                                                                                                                                                   | Volunteer answers or write on mini whiteboards. Justify reasoning: "Because the core is..." Self-check against revealed answers.                                        |

| Time / Slides             | Teacher Actions                                                                                                                                                                                                                                                                                                                                                                                                          | Student Actions                                                                                                     |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| 50–56 min<br>Slides 17–18 | <b>Growing Patterns &amp; Number Patterns:</b><br>Show Slide 17—Growing patterns get bigger by the same amount each time. Examples: ladoos stacking (1, 2, 3, 4...), Arjun's piggy bank (save 2 coins daily). Show Slide 18—Skip counting: by 2s (2, 4, 6, 8, 10), by 5s (5, 10, 15, 20, 25). Count cow legs: 4, 8, 12, 16... Count rupee coins: 5, 10, 15... Use the 100-chart; students colour every 2nd / 5th square. | Spot the 'add the same amount' rule. Choral-count skip sequences. Colour 100-chart and share what pattern they see. |
| 56–60 min<br>Slides 19–21 | <b>Indian Pattern Hunt + Wrap-Up:</b> Show Slide 19—Pattern Hunt list: rangoli, Kantha embroidery, temple gopuram, handloom fabric, market stalls. Homework challenge: find 2 patterns at home and draw them. Lead the class chant (Slide 20). Award 'Pattern Detective Badge' (Slide 21) - sticker or stamp.                                                                                                            | Contribute one Indian pattern example. Note homework assignment. Chant along; celebrate learning.                   |

## 4. Key Vocabulary

| Term                     | Definition                                                       |
|--------------------------|------------------------------------------------------------------|
| <b>Pattern</b>           | A design or sequence that repeats in the same order.             |
| <b>Pattern Core</b>      | The smallest repeating unit of a pattern.                        |
| <b>Repeating Pattern</b> | A pattern where the core keeps coming back again and again.      |
| <b>Growing Pattern</b>   | A pattern that increases by the same amount each time.           |
| <b>Extend</b>            | To continue a pattern by adding more elements.                   |
| <b>Missing Element</b>   | A piece that is absent in the middle of a pattern.               |
| <b>Skip Counting</b>     | Counting by jumping over a fixed number each time (e.g., by 2s). |
| <b>Sequence</b>          | An ordered list of numbers or objects that follow a rule.        |

## 5. Pattern Core Types — Teacher Reference

| Core Type   | Structure                  | Example Sequence                                                                   | Indian Cultural Link                                 |
|-------------|----------------------------|------------------------------------------------------------------------------------|------------------------------------------------------|
| <b>AB</b>   | 2 alternating elements     |   | Two-colour bangle stack; Red-Green traffic light     |
| <b>AAB</b>  | 2 of A, then 1 of B        |   | Kantha embroidery: two flower stitches then one leaf |
| <b>ABB</b>  | 1 of A, then 2 of B        |   | Beaded necklace: 1 red, 2 blue, 1 red, 2 blue...     |
| <b>ABC</b>  | 3 different elements cycle |   | Traffic lights: red-yellow-green cycle               |
| <b>AABB</b> | 2 of A, then 2 of B        |  | Phulkari border: two red then two yellow flowers     |
| <b>AAA</b>  | All same (trivial)         |  | String of same-colour diyas in a row                 |

## 6. Differentiation Strategies

| Learner Group            | Strategy                                                      | Activity Modification                                                                                    |
|--------------------------|---------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| <b>Emerging Learners</b> | Concrete-first approach; use blocks and beads before symbols. | Pre-cut pattern strips; student matches pieces rather than drawing. AB only.                             |
| <b>On-Track Learners</b> | Use the standard lesson flow with worksheet tasks.            | Identify core, extend by 4 elements, find 1 missing piece.                                               |
| <b>Advanced Learners</b> | Challenge with 4-element cores and growing patterns.          | Create their own ABC/AABB pattern; write a 'rule card' describing it. Extend number pattern to 10 terms. |
| <b>ELL / Bilingual</b>   | Use Hindi/Tamil vocabulary labels on visuals.                 | Pair with a bilingual buddy; accept answers in home language.                                            |

## 7. Assessment for Learning

### Formative (During Lesson)

- Thumbs-up/thumbs-down quick checks after each mini-quiz slide.
- Circulate during pair work and observe whether students identify the core before extending.

- Listen for correct use of vocabulary: 'pattern', 'core', 'repeats'.
- Exit ticket (last 3 minutes): draw a pattern with at least 2 full repetitions and label the core.

## Summative / Homework

- Indian Pattern Hunt: find and sketch 2 patterns at home; name the core type.
- Worksheet: 5 extend-the-pattern and 3 find-the-missing-element questions.
- Optional: bring a photo of a pattern found in clothing or household objects.

## Success Criteria Checklist

| Success Criterion                                  | Student Self-Check       | Teacher Observation            |
|----------------------------------------------------|--------------------------|--------------------------------|
| Can define 'pattern' in own words                  | <input type="checkbox"/> | Emerging / Developing / Secure |
| Can identify the pattern core                      | <input type="checkbox"/> | Emerging / Developing / Secure |
| Can name the core type (AB, AAB, etc.)             | <input type="checkbox"/> | Emerging / Developing / Secure |
| Can extend a repeating pattern by 2 elements       | <input type="checkbox"/> | Emerging / Developing / Secure |
| Can find a missing element                         | <input type="checkbox"/> | Emerging / Developing / Secure |
| Can skip-count by 2s and 5s to 30                  | <input type="checkbox"/> | Emerging / Developing / Secure |
| Can connect patterns to cultural/everyday examples | <input type="checkbox"/> | Emerging / Developing / Secure |

## 8. Teacher Tips & Common Misconceptions

### Common Student Misconceptions

- **Any arrangement is a pattern.** → Correct by showing a random arrangement vs. a true repeating sequence. Ask: "Does this keep repeating the same rule?"
- **Only colour makes a pattern.** → Show size patterns (big-small) and shape patterns (circle-square) to broaden understanding.
- **Confusing the full sequence with the core.** → Use brackets or underline to mark the core clearly: [●●●]●●●
- **Thinking a pattern must have exactly 2 elements.** → Demonstrate ABB, ABC, and AABB early.
- **Growing patterns are 'not real' patterns.** → Explicitly connect: "It still has a rule—it grows by the same amount each time!"

### Practical Classroom Tips

- Use body-movement patterns (clap, snap, stamp) at the start—kinesthetic learners especially benefit.
- Always identify the CORE before extending—build this as a non-negotiable habit.

- Celebrate errors as 'detective clues'—keep the atmosphere exploratory.
- The slide deck's bilingual phrases naturally invite language-mixing; encourage it.
- For bead activities, pre-thread 12 beads per pair to save time.
- When using the 100-chart for skip counting, have students use two different-coloured markers for 2s vs 5s so they can see where they overlap.
- Homework Pattern Hunt can be extended into a class gallery—students display sketches and peers guess the core type.

### CLASS CHANT (Slide 20)

**"Pattern dekho, core nikalo, pattern banao!"**

(Look at the pattern, find the core, make the pattern! Repeat 3 times, getting louder each time with claps.)

## 9. Student Worksheet Tasks (Print-Ready)

*Teacher note: Print one copy per student. Encourage students to draw or colour shapes.*

### Task A — What Comes Next? (Extend the Pattern)

1.       \_\_\_\_\_

2.       \_\_\_\_\_

3. Small diya, Big diya, Small diya, Big diya, \_\_\_\_\_

4. 2 - 4 - 6 - 8 - \_\_\_\_\_ - \_\_\_\_\_ - \_\_\_\_\_

5. 5 - 10 - 15 - \_\_\_\_\_ - \_\_\_\_\_ - \_\_\_\_\_

### Task B — Find the Missing Piece

1.     \_\_\_\_\_    

2. Red, Blue, Green, Red, \_\_\_\_\_, Green, Red, Blue, Green

3. ★ ★ ⚡ ★ ★ ⚡ ★ \_\_\_\_\_ ⚡

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4. 3 - 6 - 9 - \_\_\_\_\_ - 15 (skip count by 3s)

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### Task C — Identify & Name the Pattern Core

1. ● ● ● ● ● ● ● ● ● ●

Core = \_\_\_\_\_ Type = \_\_\_\_\_

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2. ● ● ● ● ● ● ● ● ● ●

Core = \_\_\_\_\_ Type = \_\_\_\_\_

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## Task D — Create Your Own Pattern

Draw your own pattern using shapes or colours. Make sure it repeats at least 3 times.

1. Circle your pattern core.
2. Write the core type: **AB / AAB / ABB / ABC / AABB**
3. Give your pattern a name! (e.g., 'My Rangoli Pattern')

Name: \_\_\_\_\_

## 10. Extension Activities & Homework

### In-Class Extensions (for early finishers)

- **Pattern Gallery Walk:** Post 6 patterns around the room; students circulate with a clipboard and record the core type for each.
- **Human Pattern:** 6–8 students stand in an AB or AAB sequence using colour cards; classmates identify and extend.
- **Number Pattern Challenge:** Given the rule 'add 3', can you continue the pattern from 3 to 30?
- **Design a Kolam:** Use dot paper to create a symmetrical repeating kolam pattern.

### Homework (assigned at end of lesson)

- **Indian Pattern Hunt:** Find 2 patterns at home (clothing, floor tiles, kitchen items, garden). Sketch each and write: (a) what repeats, (b) the core, and (c) the core type.
- **Number Pattern:** Starting from 10, skip count by 2s. Write the first 10 numbers in the sequence.
- **Optional Creative:** Use rangoli colours or crayons to create a beautiful border pattern for a greeting card.

## 11. Slide-by-Slide Reference Summary

| Slide(s) | Title                             | Teaching Focus                       |
|----------|-----------------------------------|--------------------------------------|
| 1        | Cover — Patterns & Sequences      | Hook with rangoli image; set context |
| 2        | Look! A Pattern Repeats!          | Introduce the idea: patterns repeat  |
| 3        | What Is a Pattern?                | Definition + 3 everyday examples     |
| 4        | Repeating Patterns & Pattern Core | Name and explain the pattern core    |
| 5        | Types of Pattern Cores            | AB, AAB, ABB, ABC, AABB with symbols |

| Slide(s) | Title                                 | Teaching Focus                         |
|----------|---------------------------------------|----------------------------------------|
| 6        | Patterns in Everyday Life             | Real-life Indian examples              |
| 7        | Extending Patterns — What Comes Next? | 3-step strategy; diya example          |
| 8        | Extending Patterns (Practice)         | Peacock-flower pair activity           |
| 9        | Missing Elements in a Pattern         | Look before & after the blank strategy |
| 10       | Pattern Detective Priya               | Bead necklace missing-element story    |
| 11–12    | Mini Quiz Q1 / Answer 1               | Identify core: flower + leaf           |
| 13–14    | Mini Quiz Q2 / Answer 2               | Extend sequence challenges             |
| 15–16    | Mini Quiz Q3 / Answer 3               | Missing bead in colour sequence        |
| 17       | Growing Patterns                      | Patterns that grow by same amount      |
| 18       | Patterns in Numbers                   | Skip counting by 2s and 5s             |
| 19       | Indian Pattern Hunt                   | Rangoli, Kantha, gopuram, handloom     |
| 20       | Recap & Chant                         | Summary + class chant                  |
| 21       | Pattern Detective Badge Earned!       | Celebration + preview next lesson      |
| 22       | Thank You — PLANRS Academy            | Closing slide                          |

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